

PATAN MULTIPLE CAMPUS

PATAN DHOKA, LALITPUR

(Department of Computer Science and Information Technology)



SOFTWARE PROJECT MANAGEMENT LAB REPORT

Submitted By:

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S.N.	Lab Task	Date	Signature
1	Student Profile Management System - Project Scheduling Consider the Student Profile Management project details with different activities, durations, predecessors, and resources. Create the project in ProjectLibre, build the WBS and RBS, generate both basic and relationship-based Gantt charts, produce Task Usage and Resource Usage reports, and draw the PERT/CPM network diagram. Identify the critical path and determine the project duration.	July 8, 2026	
2	PERT Diagram and Critical Path Given a set of project activities with predecessors and durations, draw the PERT network diagram. Perform forward pass (ES, EF) and backward pass (LS, LF) computations, calculate slack for each activity, and determine the critical path and total project duration.	July 8, 2026	
3	AOA Network Diagram and Critical Path Calculation Given a set of activities in Activity-on-Arrow (AOA) notation with node pairs and durations, compute earliest and latest event times (TE and TL), identify critical nodes and activities, and determine the critical path and total project duration.	July 8, 2026	

Objective

The objective of this lab is to understand the fundamentals of project scheduling and network modeling used in software project management. Specifically, this lab covers:

- 1.To **create a project schedule** using project management software (MS Project / OpenProj / ProjectLibre).
- 2.To **construct a WBS and RBS** - Work Breakdown Structure and Resource Breakdown Structure for the Student Profile Management project.
- 3.To **generate Gantt charts** - both basic and relationship-based (with dependency arrows).
- 4.To **produce reports** - Task Usage and Resource Usage reports with resource allocation details.
- 5.To **draw PERT/CPM diagrams** - identifying the critical path and computing floats for project network analysis.

Task 1: Student Profile Management System - Project Scheduling

Overview

This task involves creating a complete project schedule for a *Student Profile Management System* using ProjectLibre, an open-source alternative to Microsoft Project. The exercise covers project initialization, work breakdown structure (WBS) creation, Gantt chart generation, resource allocation, and network analysis.

Project Initialization

The project was created in ProjectLibre by selecting **File** → **New** and entering the following parameters:

- Project Name:** Student Profile Management
- Start Date:** 20/02/2024
- Manager:** Samir Paudel
- Organization:** Patan Multiple Campus

The project calendar was set to **Standard** with 8 working hours per day.

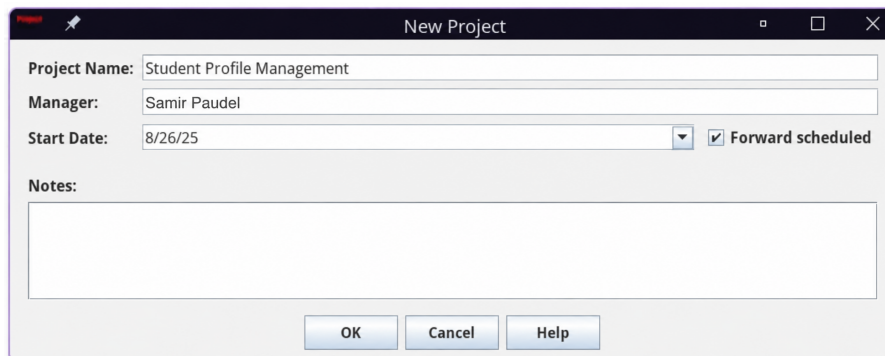


Figure 1 New Project dialog with *Student Profile Management*

Work Breakdown Structure

The project was decomposed into six sequential activities. Each activity was assigned a duration, start date, predecessor(s), and resource allocation as listed in Table 1.

S.N.	Activity	Dur.	Start	Finish	Pre.	Resource
1	Planning & Analysis	7	2024-02-20	2024-02-26	-	Team Leader (100%)
2	Requirement Collection & Definition	15	2024-02-27	2024-03-12	1	System Analyst (20%)
3	Designing	7	2024-03-13	2024-03-19	2	Designer (15%)
4	Development	21	2024-03-20	2024-04-09	2, 3	Programmer (70%)
5	Testing	8	2024-04-10	2024-04-17	2, 3, 4	Test Engineer (20%)
6	Deployment	7	2024-04-18	2024-04-24	3, 4, 5	-

Table 1 WBS Data Table as entered in ProjectLibre

All dependencies were set to Finish-to-Start (FS), meaning each successor task begins the day after the latest-finishing predecessor completes. The total project duration spans 65 calendar days, from 20 February 2024 to 24 April 2024.

Gantt Chart Visualization

With all tasks, durations, and dependencies entered, ProjectLibre automatically renders the Gantt chart for the project schedule.

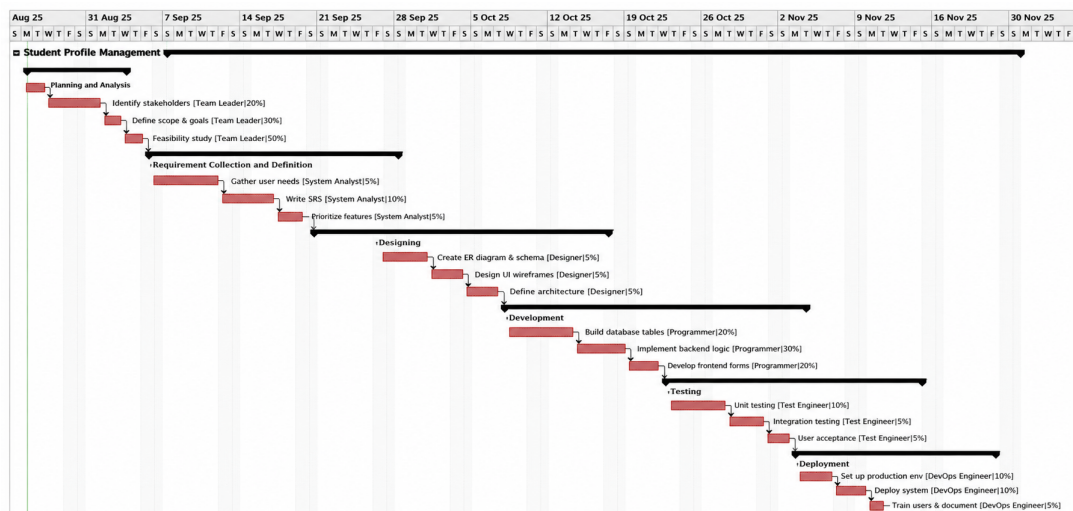


Figure 2 Gantt Chart - Student Profile Management System

WBS and RBS Diagrams

The *Work Breakdown Structure* decomposes the project into six hierarchical phases, each representing a distinct deliverable. The *Resource Breakdown Structure* categorizes resources by type-in this case, all human resources assigned to the project.

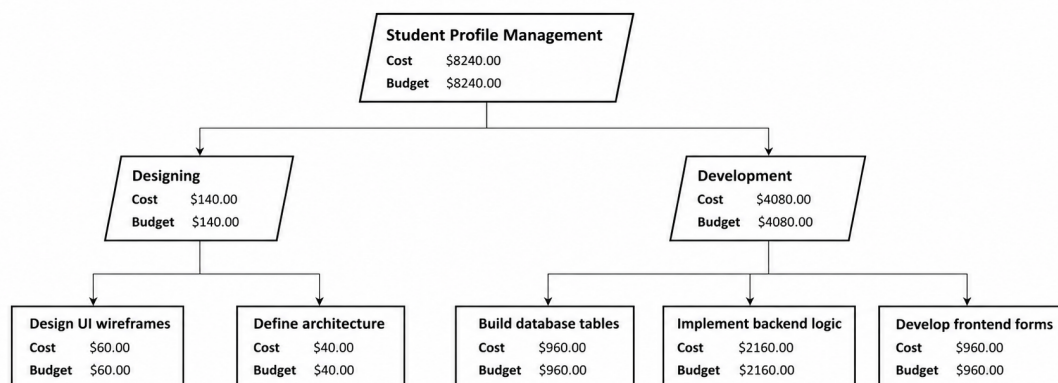


Figure 3 Work Breakdown Structure - Student Profile Management System

Team Leader Cost \$1840.00 Budget \$1840.00	System Analyst Cost \$1600.00 Budget \$1600.00	Designer Cost \$140.00 Budget \$140.00	Programmer Cost \$4080.00 Budget \$4080.00	Test Engineer Cost \$220.00 Budget \$220.00	DevOps Engineer Cost \$360.00 Budget \$360.00
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Figure 4 Resource Breakdown Structure - Student Profile Management System

Task Usage and Resource Usage Reports

Task usage and resource usage reports quantify the effort distribution. Effort is calculated as:

$$\text{Work} = \text{Duration} \times \text{Resource Units}$$

For example, the Programmer assigned at 70% for 21 days yields $21 \times 8 \times 0.70 = 117.6$ hours.

S.N.	Task	Du.	Resource	Units	Work (hrs)
1	Planning & Analysis	7d	Team Leader	100%	56.0
2	Requirement Collection & Definition	15d	System Analyst	20%	24.0
3	Designing	7d	Designer	15%	8.4
4	Development	21d	Programmer	70%	117.6
5	Testing	8d	Test Engineer	20%	12.8
6	Deployment	7d	Not assigned	-	0.0

Table 2 Task Usage Report

S.N.	Resource	Assigned Task(s)	Total Work (hrs)
1	Team Leader	Planning & Analysis	56.0
2	System Analyst	Requirement Collection & Definition	24.0
3	Designer	Designing	8.4
4	Programmer	Development	117.6
5	Test Engineer	Testing	12.8

Table 3 Resource Usage Report

Network Modeling - PERT Chart and CPM

The PERT/CPM network diagram was generated using ProjectLibre's **View** → **Network Diagram** option. Each node displays the task ID, name, duration, and the computed Early Start (ES), Early Finish (EF), Late Start (LS), and Late Finish (LF). Critical tasks-those with zero slack-are highlighted in red.

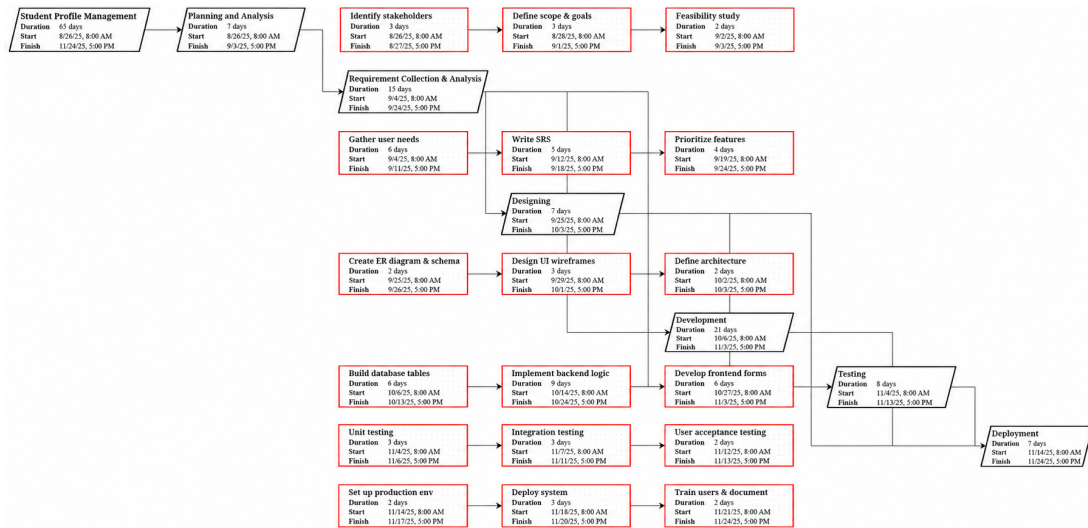


Figure 5 PERT/CPM Network Diagram (AON) - Student Profile Management System

Analysis

Since every activity depends on the completion of all its predecessors in a purely sequential chain ($1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$), there exists only one path through the network. Consequently, all six activities are critical with zero slack.

Metric	Result
Critical Path	$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$
Project Duration	$7 + 15 + 7 + 21 + 8 + 7 = \mathbf{65 \text{ days}}$

Task 2: PERT Diagram and Critical Path

Problem Definition

A project consists of six activities with the precedence relationships and durations given in Table 4.

Activity	Predecessor(s)	Duration (days)
A	-	2
B	A	4
C	A	5
D	B	6
E	C	3
F	D, E	4

Table 4 Activities with Predecessors and Durations

Forward Pass

The forward pass computes the Earliest Start (ES) and Earliest Finish (EF) times for each activity:

$$ES(A) = 0, \quad EF(A) = 0 + 2 = 2$$

$$ES(B) = EF(A) = 2, \quad EF(B) = 2 + 4 = 6$$

$$ES(C) = EF(A) = 2, \quad EF(C) = 2 + 5 = 7$$

$$ES(D) = EF(B) = 6, \quad EF(D) = 6 + 6 = 12$$

$$ES(E) = EF(C) = 7, \quad EF(E) = 7 + 3 = 10$$

$$ES(F) = \max(EF(D), EF(E)) = \max(12, 10) = 12, \quad EF(F) = 12 + 4 = 16$$

Backward Pass

The backward pass computes the Latest Finish (LF) and Latest Start (LS) times:

$$LF(F) = 16, \quad LS(F) = 16 - 4 = 12$$

$$LF(D) = LS(F) = 12, \quad LS(D) = 12 - 6 = 6$$

$$LF(E) = LS(F) = 12, \quad LS(E) = 12 - 3 = 9$$

$$LF(B) = LS(D) = 6, \quad LS(B) = 6 - 4 = 2$$

$$LF(C) = LS(E) = 9, \quad LS(C) = 9 - 5 = 4$$

$$LF(A) = \min(LS(B), LS(C)) = \min(2, 4) = 2, \quad LS(A) = 2 - 2 = 0$$

Slack Computation and Critical Path

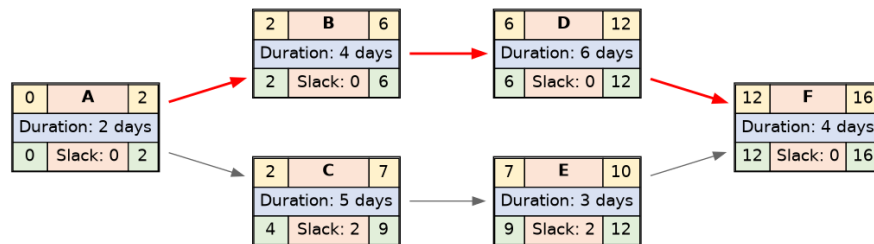
Slack is calculated as $Slack = LS - ES$. Activities with zero slack lie on the critical path.

Path Comparison

Two paths emerge from the network:

Activity	ES	EF	LS	LF	Slack	Critical
A	0	2	0	2	0	Yes
B	2	6	2	6	0	Yes
C	2	7	4	9	2	No
D	6	12	6	12	0	Yes
E	7	10	9	12	2	No
F	12	16	12	16	0	Yes

Table 5 Slack / Float Analysis



PERT/CPM Network Diagram - Critical Path: A-B-D-F , Project Duration = 16 days

Figure 6 PERT Network Diagram - Activities A to F

Path A-B-D-F = 2 + 4 + 6 + 4 = 16 days

Path A-C-E-F = 2 + 5 + 3 + 4 = 14 days

The critical path is the longest path through the network.

Metric	Result
Critical Path	A → B → D → F
Project Duration	16 days

Task 3: AOA Network Diagram and Critical Path Calculation

Problem Definition

A project is represented using Activity-on-Arrow (AOA) notation, where each activity is defined by its start and end node indices (i, j) with associated durations, as shown in Table 6.

Activity (i, j)	Duration (Days)
(1, 2)	4
(1, 3)	6
(2, 4)	5
(3, 4)	3
(4, 5)	2
(3, 5)	8

Table 6 Project activities in AOA notation

Forward Pass - Earliest Event Times

The earliest event time $E(j)$ for each node is computed starting from node 1:

$$E(1) = 0$$

$$E(2) = E(1) + 4 = 4$$

$$E(3) = E(1) + 6 = 6$$

$$E(4) = \max[E(2) + 5, E(3) + 3] = \max[9, 9] = 9$$

$$E(5) = \max[E(4) + 2, E(3) + 8] = \max[11, 14] = 14$$

Backward Pass - Latest Event Times

The latest event time $L(i)$ for each node is computed backward from node 5:

$$L(5) = 14$$

$$L(4) = L(5) - 2 = 12$$

$$L(3) = \min[L(4) - 3, L(5) - 8] = \min[9, 6] = 6$$

$$L(2) = L(4) - 5 = 7$$

$$L(1) = \min[L(2) - 4, L(3) - 6] = \min[3, 0] = 0$$

Event Time Table

Critical Path Determination

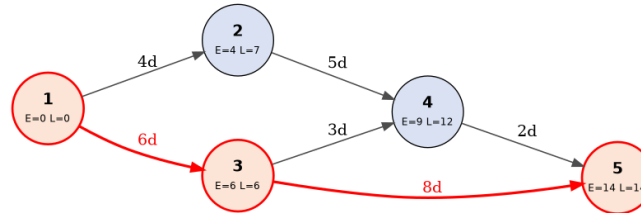
An activity (i, j) lies on the critical path if and only if:

$$E(i) = L(i), \quad E(j) = L(j), \quad E(i) + d_{ij} = E(j)$$

Applying this condition:

Node	Earliest E	Latest L	Slack	Critical
1	0	0	0	Yes
2	4	7	3	No
3	6	6	0	Yes
4	9	12	3	No
5	14	14	0	Yes

Table 7 Event time analysis



AOA (Activity-on-Arrow) Network Diagram - Critical Path: 1-3-5 (via activities (1,3)+(3,5)), Project Duration = 14 days

Figure 7 AOA Network Diagram - Nodes 1 to 5

(1, 3) : $E(1) + 6 = 6 = E(3)$, both nodes critical → **Critical**

(3, 5) : $E(3) + 8 = 14 = E(5)$, both nodes critical → **Critical**

(1, 2), (2, 4), (3, 4), (4, 5) : fail the equality test → Not critical

Metric	Result
Critical Path	1 → 3 → 5 (via activities (1,3) and (3,5))
Project Duration	6 + 8 = 14 days

Conclusion

This lab provided hands-on experience in software project scheduling using a project management tool (ProjectLibre) and manual network analysis. The key learnings from this lab are summarized below.

Key Learnings

- **Project Planning** - Creating a project plan with a Work Breakdown Structure (WBS) linking tasks, durations, dates, dependencies and resources.
- **Gantt Charting** - Visualizing the schedule through basic and relationship (dependency-linked) Gantt charts.
- **WBS and RBS** - Decomposing the project into WBS and RBS hierarchies for clearer planning and resource allocation.
- **Effort Tracking** - Generating Task Usage and Resource Usage reports to track effort distribution.
- **Critical Path Analysis** - Applying the forward and backward pass technique of PERT/CPM to compute ES, EF, LS, LF, slack, and identify the critical path.

Summary of Results

S.N.	Exercise	Critical Path Duration
1	Student Profile Management (ProjectLibre)	65 days
2	PERT Network Analysis (Activities A–F)	16 days
3	AOA Network Analysis (Nodes 1–5)	14 days